

Social Data Mining of Online Discourse Around President Donald Trump

Project Group 5

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Data Source: Manch Hui 2020 US Election Dataset (Kaggle) ~1.7M Tweets



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Motivations for the Project Topic

- Politically Themed Social-Media discourse is often stated to be 'polarized', i.e., ideologically and structurally divided.
- The 2020 POTUS election was a global phenomenon because of the country's general impact on the rest of the world; one of the largest US election by sheer number of votes.
- Analysing Trump vs Biden discourse in twitter yields various insights about their respective political ideologies and communities formed around them.
- **What we could learn:** If the linguistic gap correlates strongly with structural distance, it confirms that social segregation drives ideological radicalization. If they do *not* correlate (e.g., structurally distant groups still discuss the same topics), it suggests that mass media or external events drive the conversation more than the social network structure itself.

Research Questions

1. **Topological Analysis:** Does the Trump-supporting network exhibit higher topological rigidity when compared to the Biden-supporting network, and if yes, by how much?
2. **Semantic Analysis:** How effectively can semantic divergence between Pro-Biden and Anti-Biden communities be quantified using Topic Modelling, and does the linguistic gap correlate with the structural distance between these communities?
3. **Multiplex Analysis:** Do the "Endorsement Layer" (Retweets) and "Discussion Layer" (Mentions) display negative structural correlation and low edge overlap?

Data Exploration and Preprocessing

Data Readiness & Scope: Inspect schema, confirm key fields are present and usable. Assess missingness and noise (locations, bios, duplicates, bots). Define the analysis window and sampling strategy (e.g., key week, sub-sample size).

Output: Short notes on data quality and operational constraints.

Structural Feasibility (RQ1): Build directed user-to-user graphs for retweets (endorsement) and mentions (discussion). Apply Louvain community detection.

Output: Confirmation of coherent sub-communities and assessment of network scale manageability.

Semantic & Ideological Feasibility (RQ2 & RQ3): Preprocess text: filter language, clean URLs/emojis, and handle near-duplicate retweets. Analyse linguistic variance between groups.

Output: Determination of whether pro- vs. anti- Trump communities are distinguishable and if stance labelling is practical.

Multilayer Analysis (RQ3): Treat retweet and mention graphs as two distinct layers on a shared user set. Compute edge overlap (e.g., Jaccard), find strong-tie overlaps, and compare modularity/polarization across layers.

Output: Quantification of cross-layer consistency and interaction patterns.

Methodology

1. Data Preprocessing

Parsers: Custom Regex to split RT @user (Endorsement) vs. @user (Discussion).

Filters: Remove "Bogus Rows" (parsing errors) & Bots (>50 tweets/day).

2. Structural Analysis - Louvain Community Detection on the Retweet Layer.

Goal:: Find ground-truth communities (Trump Hive vs. Biden Hive).

Key Metric: Modularity (Q) – Quantifying the density of the echo chamber.

3. Semantic Analysis - BERTopic Modelling & TF-IDF with Weak Supervision.

Goal: Map the "Semantic Wall"—distinct vocabularies separating the clusters.

Key Metric: Jensen-Shannon Divergence between community topic distributions.

4. Multiplex Dynamics - Layer Comparison (Endorsement vs. Discussion Graphs).

Goal: Test if users trust (retweet) locally but fight (mention) globally.

Key Metric: Jaccard Similarity (J) of edge overlaps between layers.

Tools: Python (Pandas, NetworkX), Gephi (Visualization), Scikit-Learn (Classification).

Key Insights (Research Question #1 - Structure)

Network Preparation:

1. **Dataset:** Uses pre-processed CSVs (Removed noise, bots, non-English tweets) of Trump and Biden
2. **Edges:**
 - Regex `RT @(\w+)`: to identify retweets and `@(\w+` (excluding the ones already identified as retweets) to identify mentions
 - Retweets are assigned a weight of 2 and Mentions are assigned a weight of 1.
 - Retweets seen as endorsements (thus a higher weight), mentions seen as generic interactions
 - Bidirectional retweets/mentions (between the same two users) are cumulative; e.g., two users retweeting each other and only one mention between them would result in a total edge weight of 5 (2+2+1) - to consider the strength of ties.
3. **Network Graph Building:** Uses `igraph` to build an undirected weighted graph.

Key Insights (Research Question #1 - Structure)

- **Insight 1a: Rigidity Gap (ΔQ):**

- Both networks are non-random ($p < 0.001$)
- Trump-centric discourse shows a 65% **larger gap** than Biden's - Trump "Bunker" is a deliberate social structure.

- **Insight 1b: Overall Topology:**

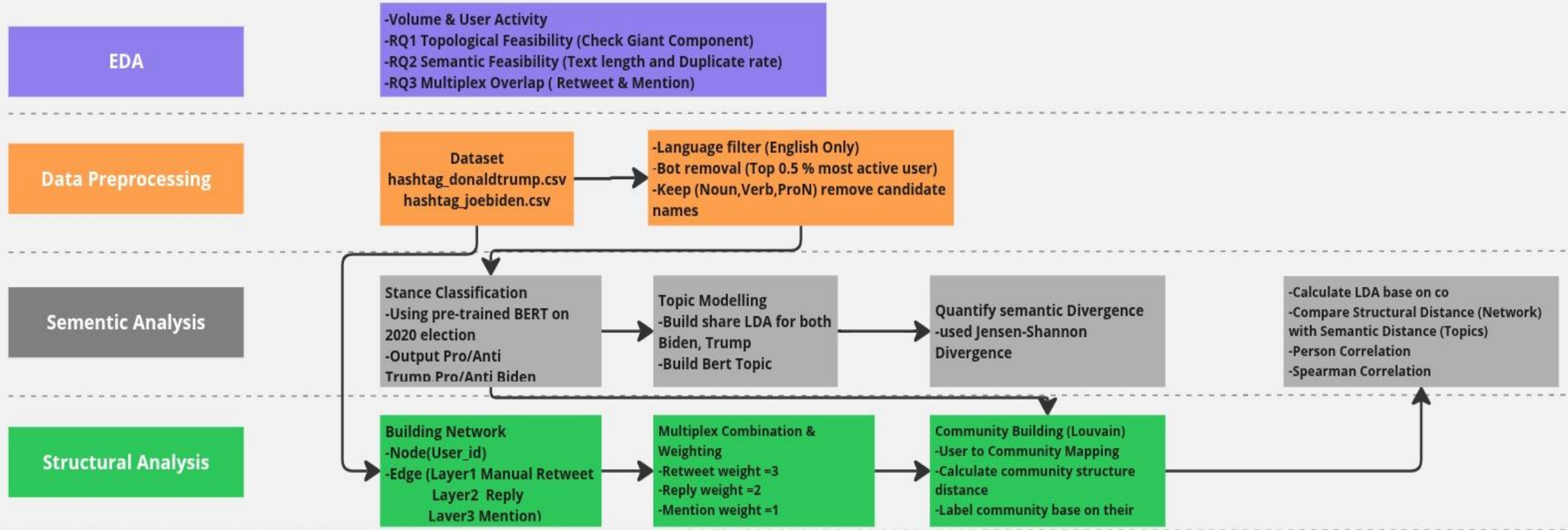
- Trump's network exhibits lower clustering and fewer silos where interaction is dictated by a few dominant hubs.
- Biden's discourse has nearly **3x higher clustering** and more fragmented silos - more interconnected and distributed mainstream coalition.

- **Insight 1c: Proportional Failure:**

- Collapses after removing only **0.11%** of the population (32 hubs), LCC shrinks by **30.16%** (High Fragility).
- Requires removing 0.56% of the population (100 hubs) to reach the same collapse threshold (High Resilience).

Key Insights (Research Question #2 - Semantic)

Research Question 2



Key Insights (Research Question #2 - Semantic)

Divergence Significance

- Pro vs Anti-Trump: JS = 0.243 (SharedLDA) / 0.265 (BERTopic) - Highest divergence
- Pro vs Anti-Biden: JS = 0.163 (SharedLDA) / 0.169 (BERTopic) - Lower divergence (~50% less)

Trump shows stronger semantic isolation driven by disjoint vocabularies (e.g., Pro-Trump Topic 4: "KAG, CCP"; Anti-Trump Topic 1: "racist, lie")

Does Linguistic Gap Correlate with Structural Distance?

Merge structural distance CSV (*weighted shortest-path distance between communities*) with semantic distance (JS divergence) on community pairs → compute Pearson & Spearman correlations

- Pearson $r = 0.361$, $p = 0.01$ (linear association)
- Spearman $r = 0.322$, $p = 0.0226$ (rank association)

Both correlations are positive & significant: greater structural distance → greater semantic divergence

Key Insights (Research Question #3 - Multilayer)

Why Study Multiple Layers?

- Political interaction on X is **not a single network** different actions reveal **different social mechanisms**
- **Retweets = agreement & trust** → who users *support*
- **Mentions = debate & confrontation** → who users *challenge*

How Networks Are Defined ?

- **Nodes** = users (- RT layer: 489 nodes, Mention layer: 208372 nodes)
- **Edges (Layer 1)** = RT @user (endorsement connections - RT layer: 291 edges)
- **Edges (Layer 2)** = @user mentions (discussion connections - Mention layer: 357041 edges)
- Same users, **different ties** → two-layer (multiplex) network

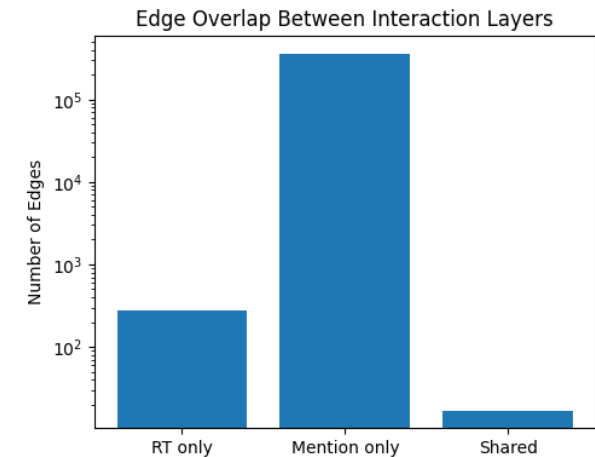
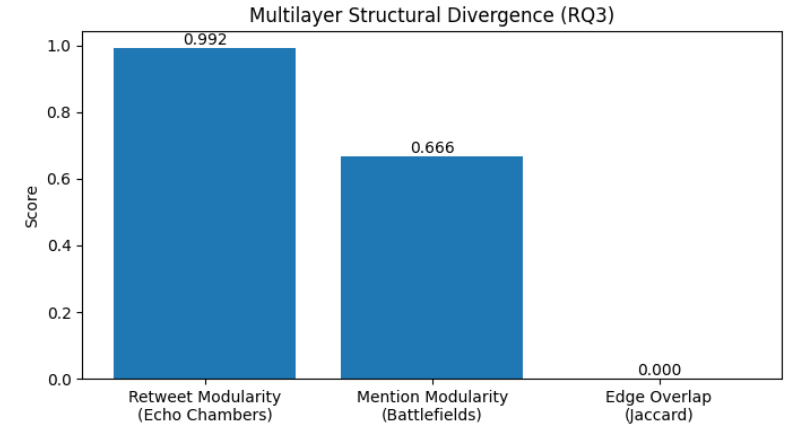
Why This Matters ?

- Looking at one layer which hides structure
- Multilayer analysis can reveal **echo chambers, conflict arenas, and hidden polarization**

Key Insights (Research Question #3 - Multilayer)

Do Retweet & Mention Layers Diverge Structurally? YES

- **Very Low Edge Overlap (< 0.1):** Users do not mention the same accounts they retweet $\rightarrow$ endorsement & conversation occur in separate spaces
- **Retweet Layer = High Modularity (Echo Chambers):** Users cluster tightly within ideologically similar groups $\rightarrow$ trust, loyalty, and reinforcement (Modularity (Retweet): 0.992)
- **Mention Layer = Lower Modularity (Battlefields):** More cross-community interaction $\rightarrow$ debate, criticism & conflict (Modularity (Mention): 0.666)
- **Near-Zero Cross-Layer Consistency:** The same user behaves differently depending on interaction type. $\rightarrow$ identities shift between supporting vs. arguing



Validity and Method Weakness

External Validity (Applicability to the target system/real world)

- Status: Not Satisfied / Limited. (The target system is often the "US Electorate," but the observed system is "Twitter users using specific hashtags.")

Construct Validity (Do data metrics match theoretical concepts?)

- Status: Satisfied. (For instance, the "Trump Pro" and "Biden Pro" found via Louvain community detection are validated by the semantic analysis)
- While automated stance detection (BERT) aligns with community structure, we did not perform manual qualitative coding to verify the nuance of these labels, leaving potential for misclassification in sarcastic or ambiguous tweets

Data Validity (Is the data a neutral/complete representation?)

- Status: Information Missing / Potential Bias. (The study relies on specific keywords (#donaldtrump, #joe Biden). This introduces selection bias, as it excludes discourse using other popular tags (e.g., #MAGA, #VoteBlue) or tweets without hashtags)

Weakness

- Arbitrary Bot Filtering: There is data-driven distribution cut 0.5 % set for bot filter that may remove hyper-active real activists while missing slower, more sophisticated bots, potentially skewing the structural analysis .
- Keyword Selection Bias: The dataset relying solely on two candidate name hashtags misses a vast portion of the election discourse, potentially underestimating the size or connectivity of the networks.
- Weighting Assumptions: We assigns arbitrary weights (Retweet=3, Reply=2 , Mention=1). While retweets generally signal endorsement, mentions are ambiguous (can be support or attack).
- Limitation of Retweet Extraction: *Our structural analysis found low number of edges. The dataset appears to have been pre-filtered to exclude native 'button-click' retweets . The small number of retweet edges we did capture were likely 'manual' retweets (where users explicitly typed 'RT @user...')*

Conclusion

The Trump "Bunker"

- Structure: Rigid, centralized star topology ($\Delta Q = +65\%$, collapses at 0.11%).
- Semantics: Near-zero correlation between structure and meaning.
- Insight: Distance is compressed; everyone hears the same echo. Uniformity replaces geography.
- Risk: Designed for control, but critically vulnerable to decapitation.

The Biden "Mesh"

- Structure: Distributed, interconnected coalition (3x clustering, robust to 0.56%).
- Semantics: Significant correlation ($r=0.36$) between structure and meaning.
- Insight: Ideological identities (progressive/moderate) are distinct but bridged. Diversity is managed, not suppressed.
- Strength: Survives through redundant connections, not central command.

Overall Conclusion: *One network demands submission to a central node; the other survives through the distributed strength of its connections.*